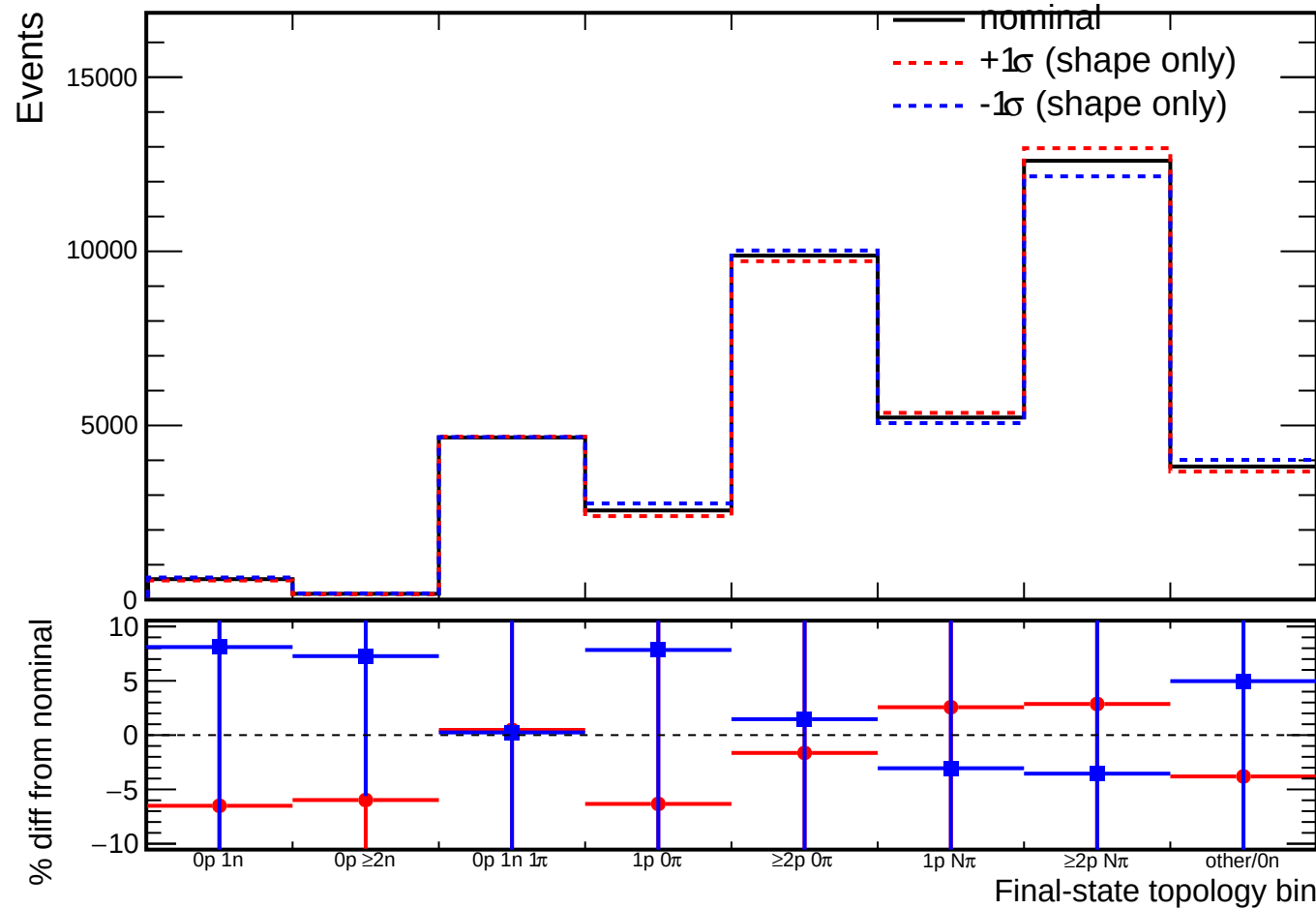
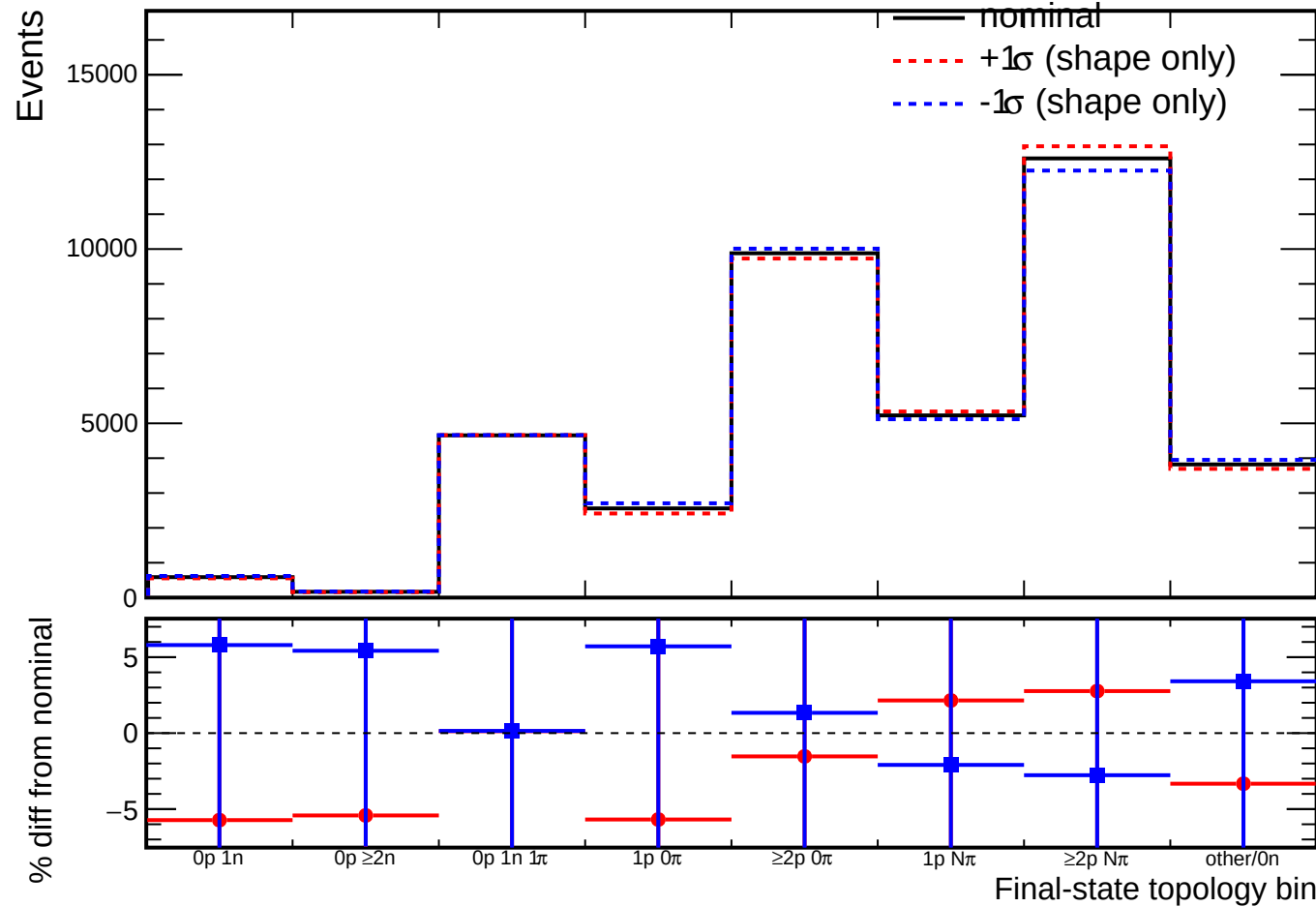


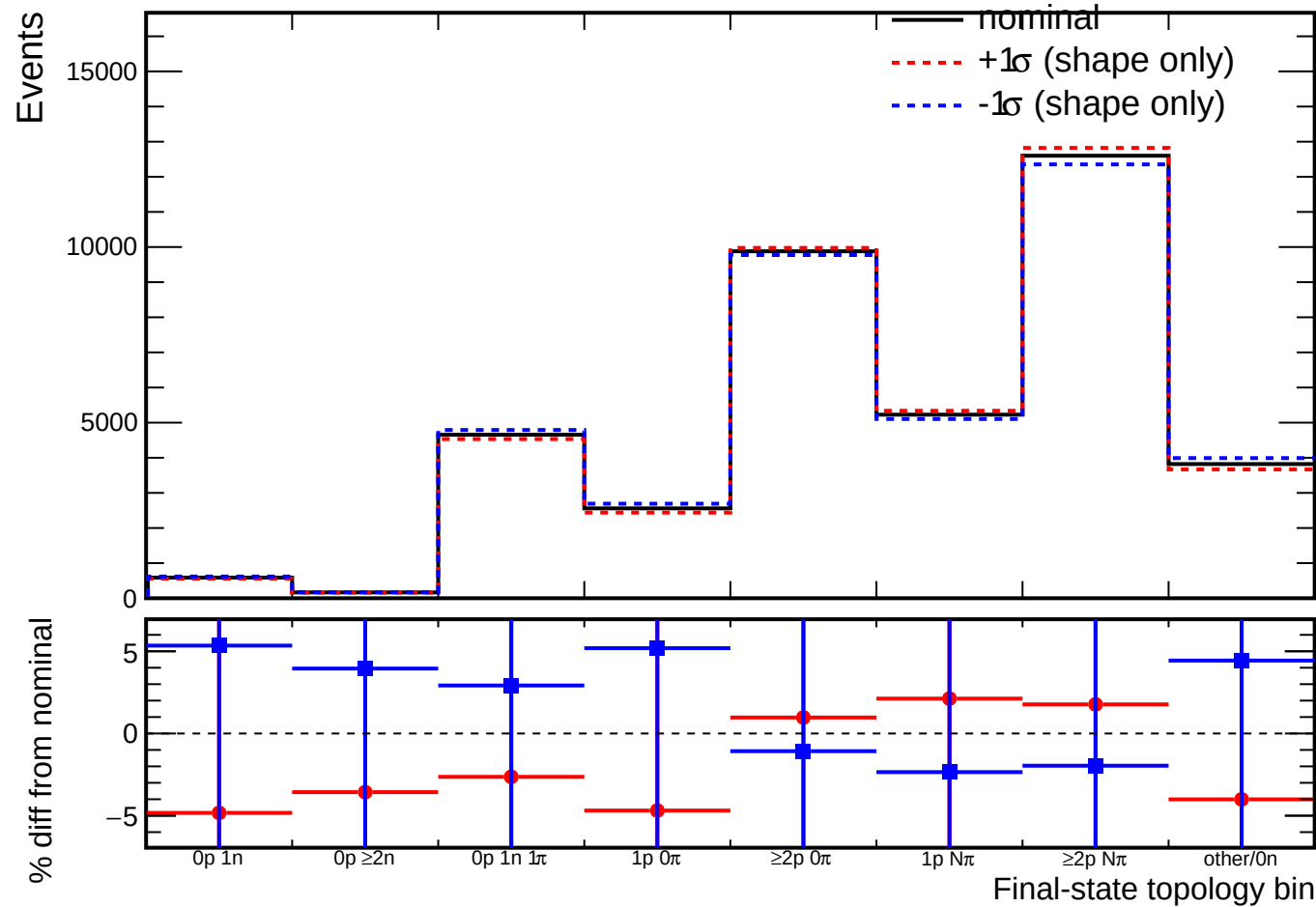
FHC [RES]: MaCCRES (shape only) $\chi^2_{\text{shape}, +1\sigma} = 35.44$ $\chi^2_{\text{shape}, -1\sigma} = 52.77$ $\Delta\langle x \rangle_{+1\sigma} = +0.0182$ $\Delta\langle x \rangle_{-1\sigma} = -0.0236$
 $\chi^2_{\text{raw}, +1\sigma} = 240.84$ $\chi^2_{\text{raw}, -1\sigma} = 272.54$ rate: +7.12% / -7.59% $\Delta\sigma^2$: -0.0382 / +0.0542



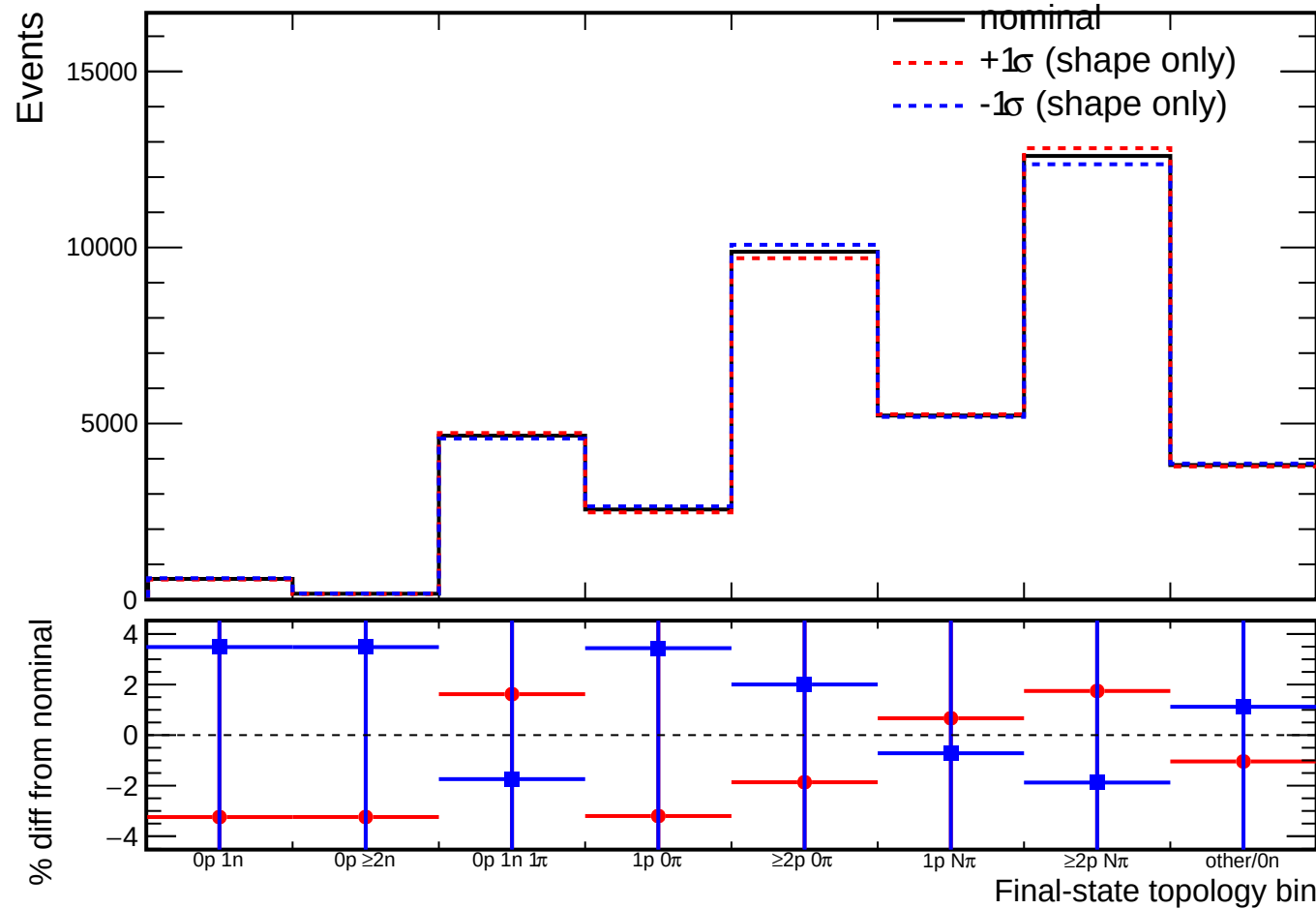
FHC [RES]: MvCCRES (shape only) $\chi^2_{\text{shape}, +1\sigma} = 29.31$ $\chi^2_{\text{shape}, -1\sigma} = 28.99$ $\Delta\langle x \rangle_{+1\sigma} = +0.0183$ $\Delta\langle x \rangle_{-1\sigma} = -0.0187$
 $\chi^2_{\text{raw}, +1\sigma} = 193.77$ $\chi^2_{\text{raw}, -1\sigma} = 152.76$ rate: +6.38% / -5.67% $\Delta\sigma^2$: -0.0351 / +0.0372



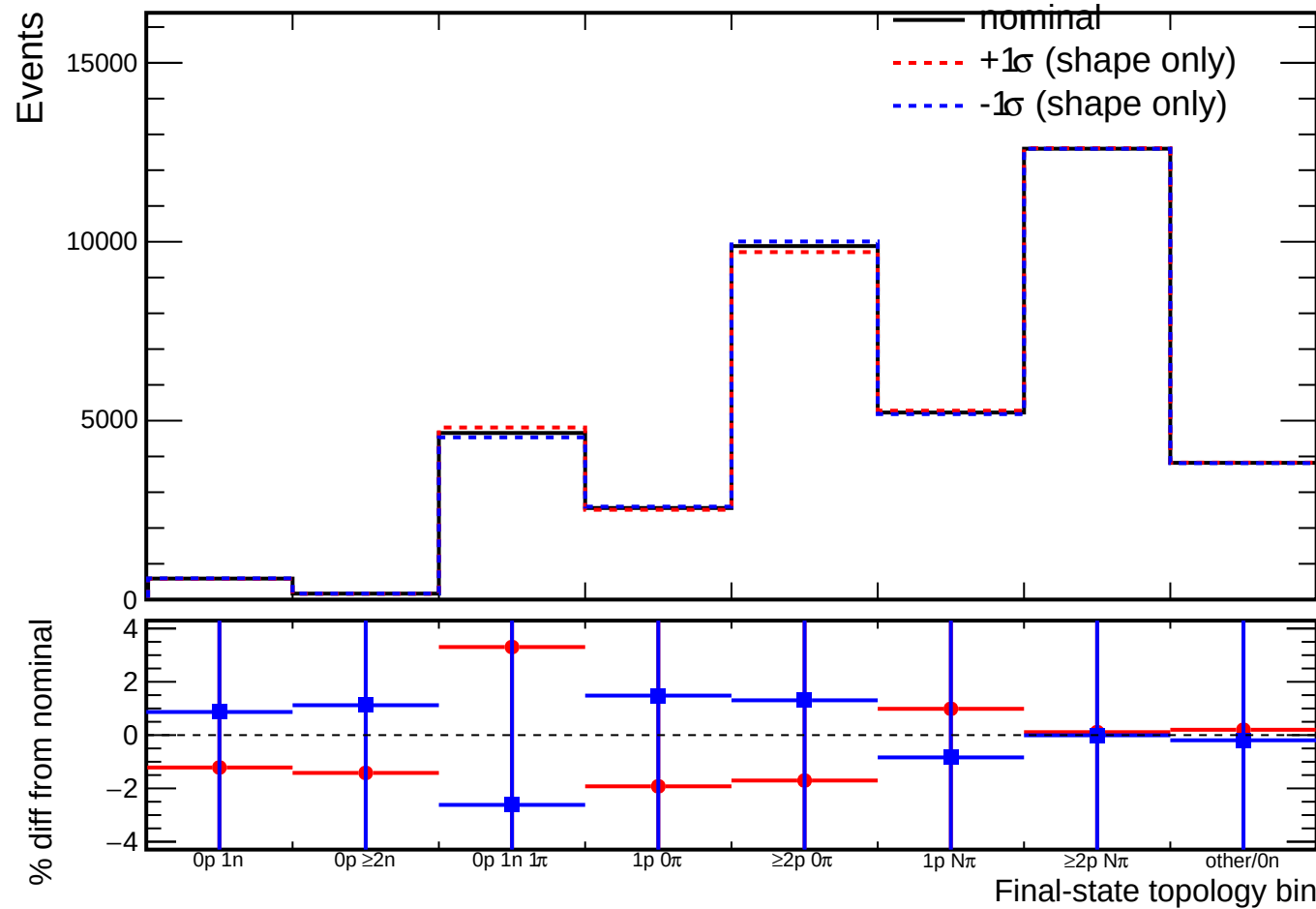
FHC [RES]: RESDeltaScaleSyst (shape only) $\chi^2_{\text{shape},+1\sigma}=23.82$ $\chi^2_{\text{shape},-1\sigma}=29.18$ $\Delta\langle x \rangle_{+1\sigma}=+0.0151$ $\Delta\langle x \rangle_{-1\sigma}=-0.0167$
 $\chi^2_{\text{raw},+1\sigma}=127.98$ $\chi^2_{\text{raw},-1\sigma}=127.98$ rate: +5.07% / -5.07% $\Delta\sigma^2$: -0.0587 / +0.0645



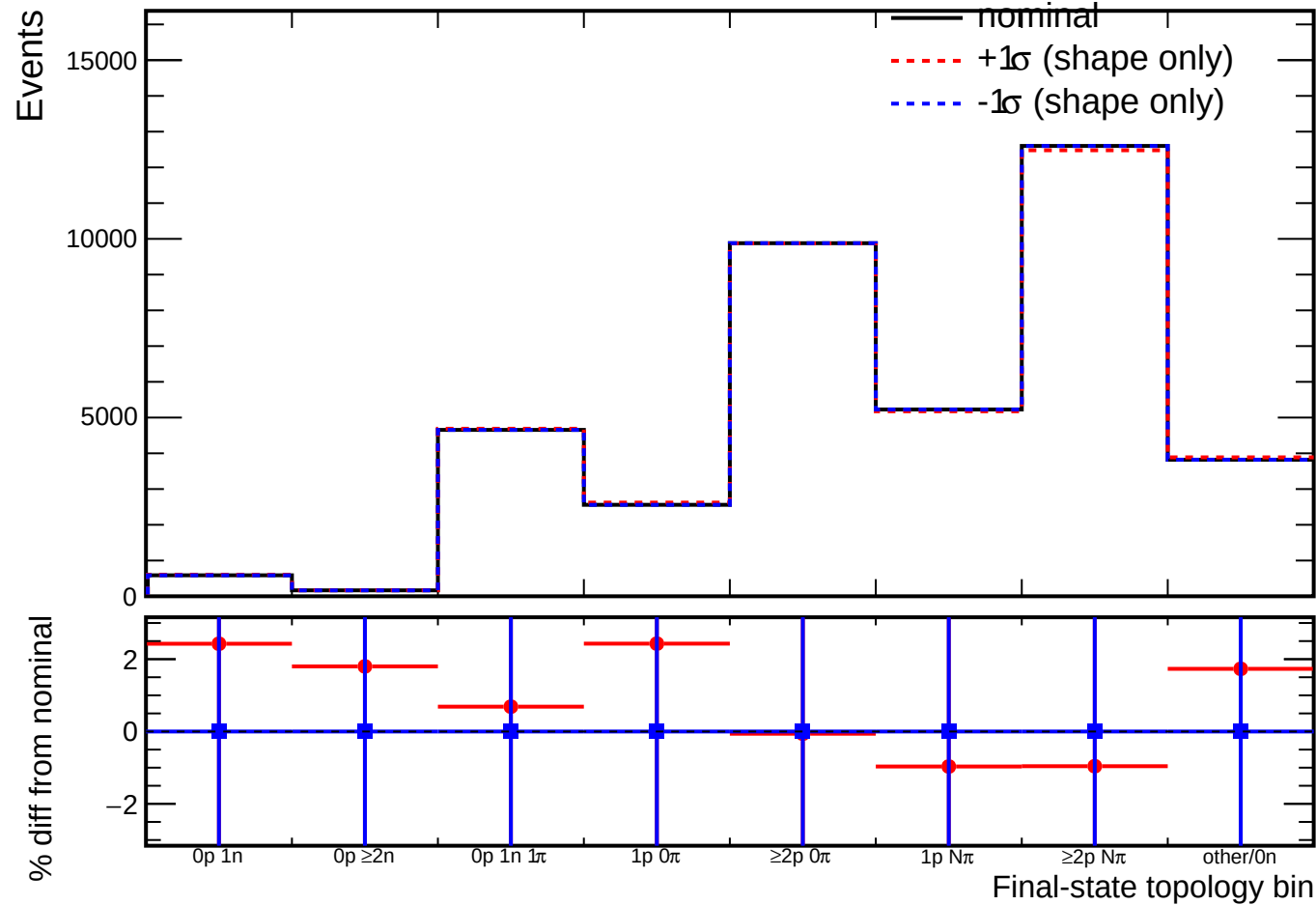
FHC [RES]: RESOtherScaleSyst (shape only) $\chi^2_{\text{shape},+1\sigma}=12.56$ $\chi^2_{\text{shape},-1\sigma}=14.50$ $\Delta\langle x \rangle_{+1\sigma}=+0.0096$ $\Delta\langle x \rangle_{-1\sigma}=-0.0103$
 $\chi^2_{\text{raw},+1\sigma}=64.59$ $\chi^2_{\text{raw},-1\sigma}=64.59$ rate: +3.60% / -3.60% $\Delta\sigma^2$: -0.0026 / +0.0026



FHC [RES]: MaNCRES (shape only) $\chi^2_{\text{shape},+1\sigma}=9.56$ $\chi^2_{\text{shape},-1\sigma}=5.88$ $\Delta\langle x\rangle_{+1\sigma}=-0.0030$ $\Delta\langle x\rangle_{-1\sigma}=+0.0028$
 $\chi^2_{\text{raw},+1\sigma}=27.39$ $\chi^2_{\text{raw},-1\sigma}=15.31$ rate: +2.10% / -1.56% $\Delta\sigma^2$: +0.0196 / -0.0157

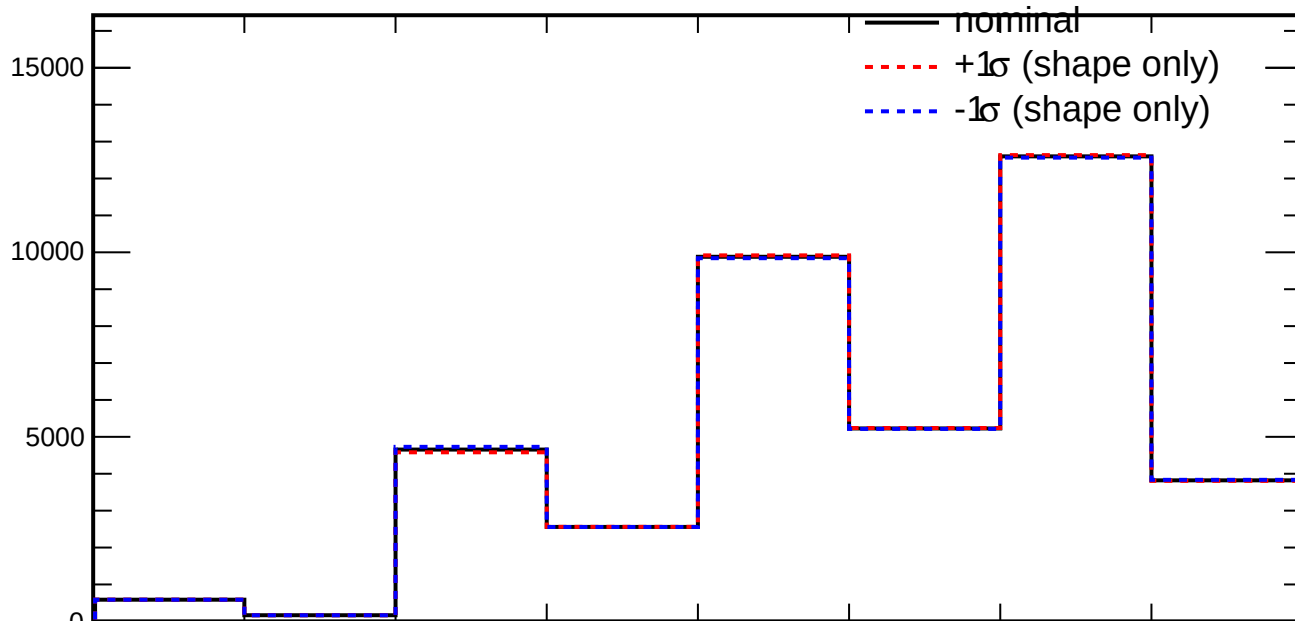


FHC [RES]: LowQ2RESSupp2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=4.93$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=-0.0072$ $\Delta\langle x\rangle_{-1\sigma}=+0.0000$
 $\chi^2_{\text{raw},+1\sigma}=28.50$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: -2.46% / +0.00% $\Delta\sigma^2$: +0.0227 / +0.0000

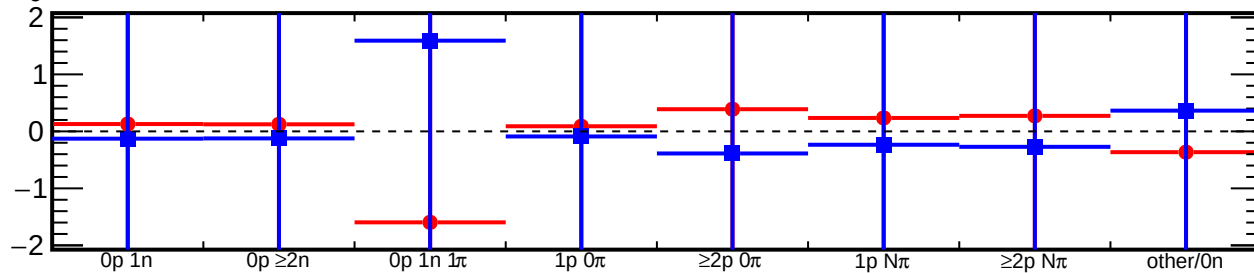


FHC [RES]: RESvpvnRatioNuXSecSyst (shape only) $\chi^2_{\text{shape}, +1\sigma} = 1.51$ $\chi^2_{\text{shape}, -1\sigma} = 1.50$ $\Delta\langle x \rangle_{+1\sigma} = +0.0046$ $\Delta\langle x \rangle_{-1\sigma} = -0.0046$
 $\chi^2_{\text{raw}, +1\sigma} = 1.60$ $\chi^2_{\text{raw}, -1\sigma} = 1.60$ rate: -0.15% / +0.15% $\Delta\sigma^2$: -0.0129 / +0.0128

Events

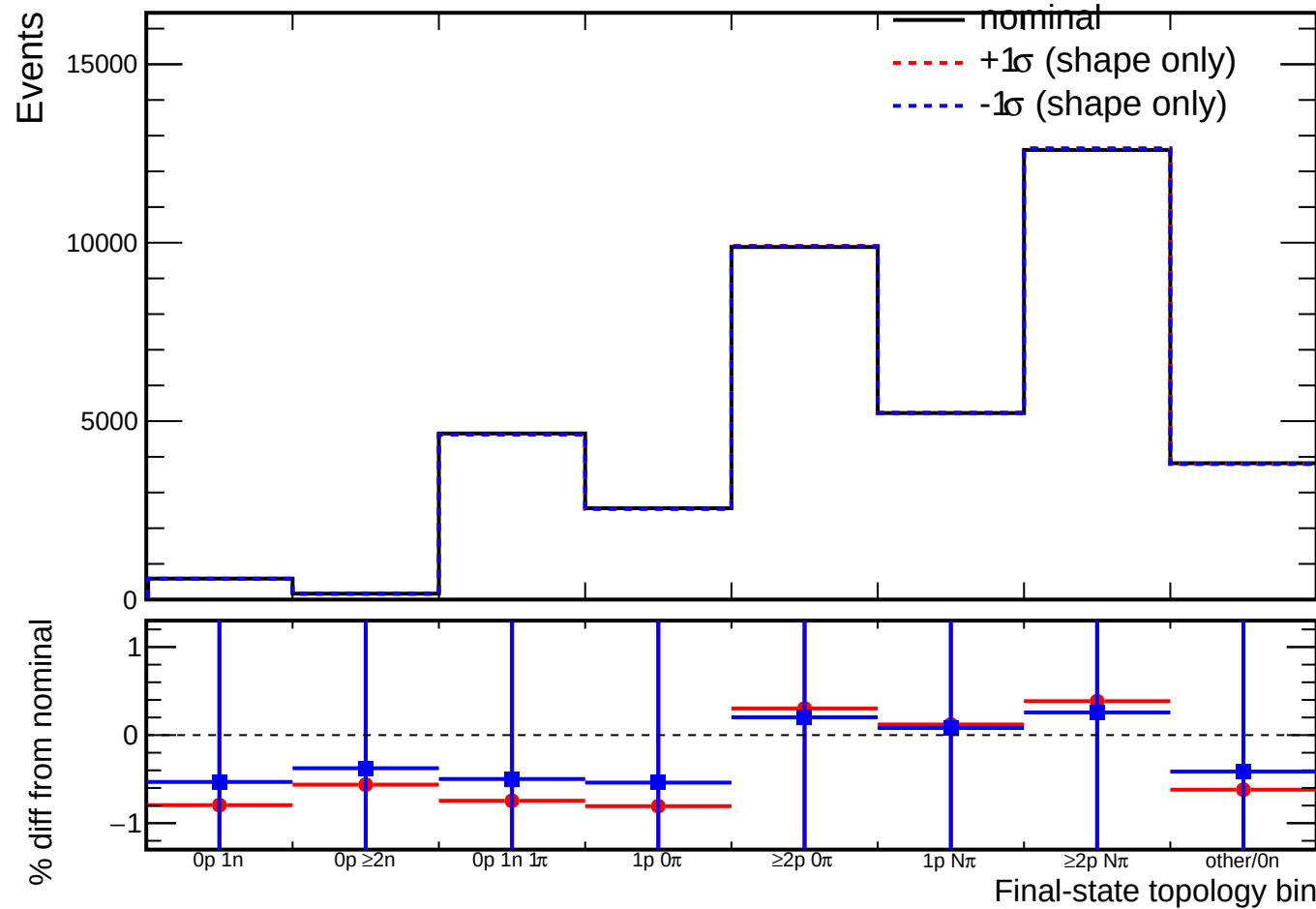


% diff from nominal

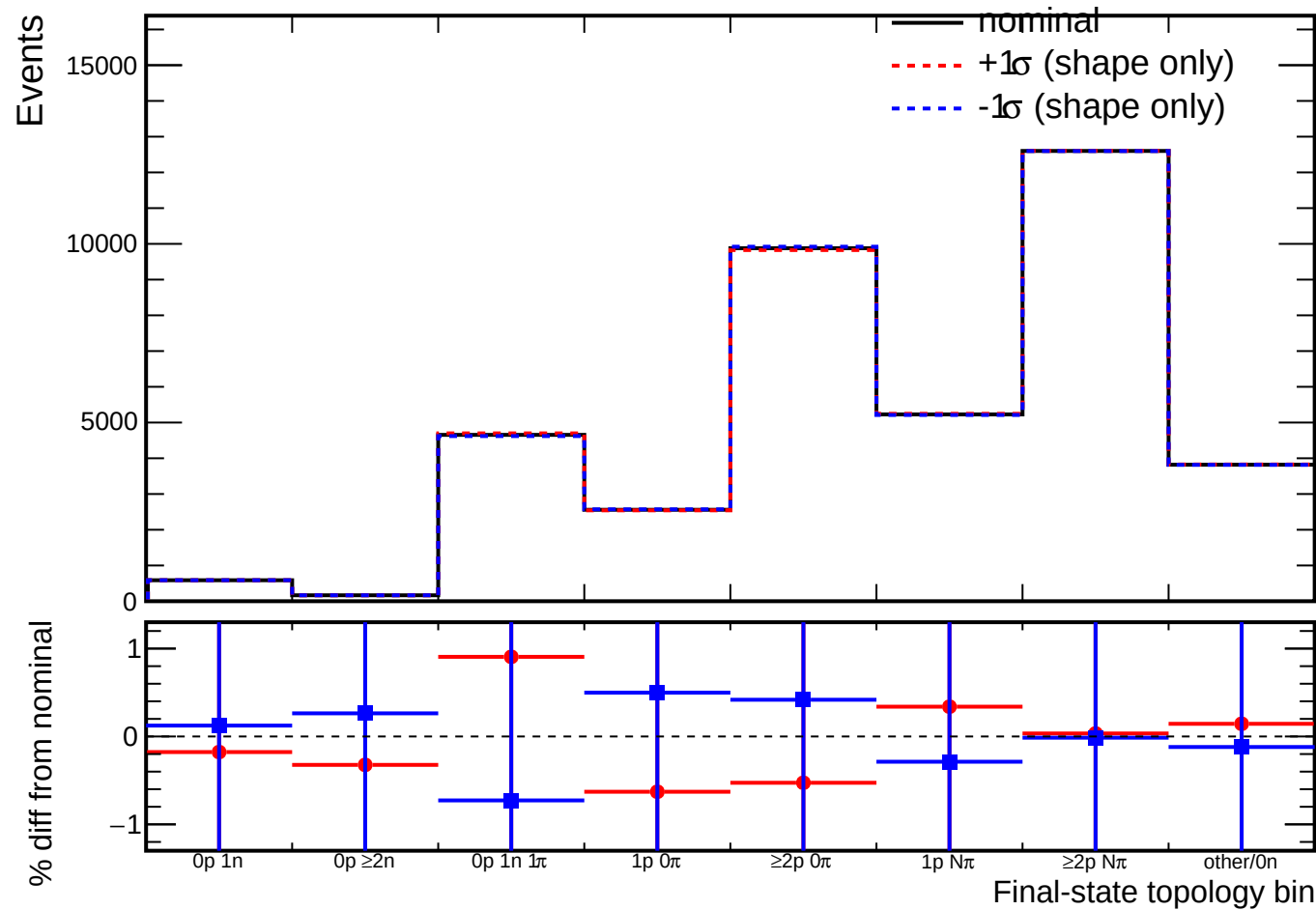


Final-state topology bin

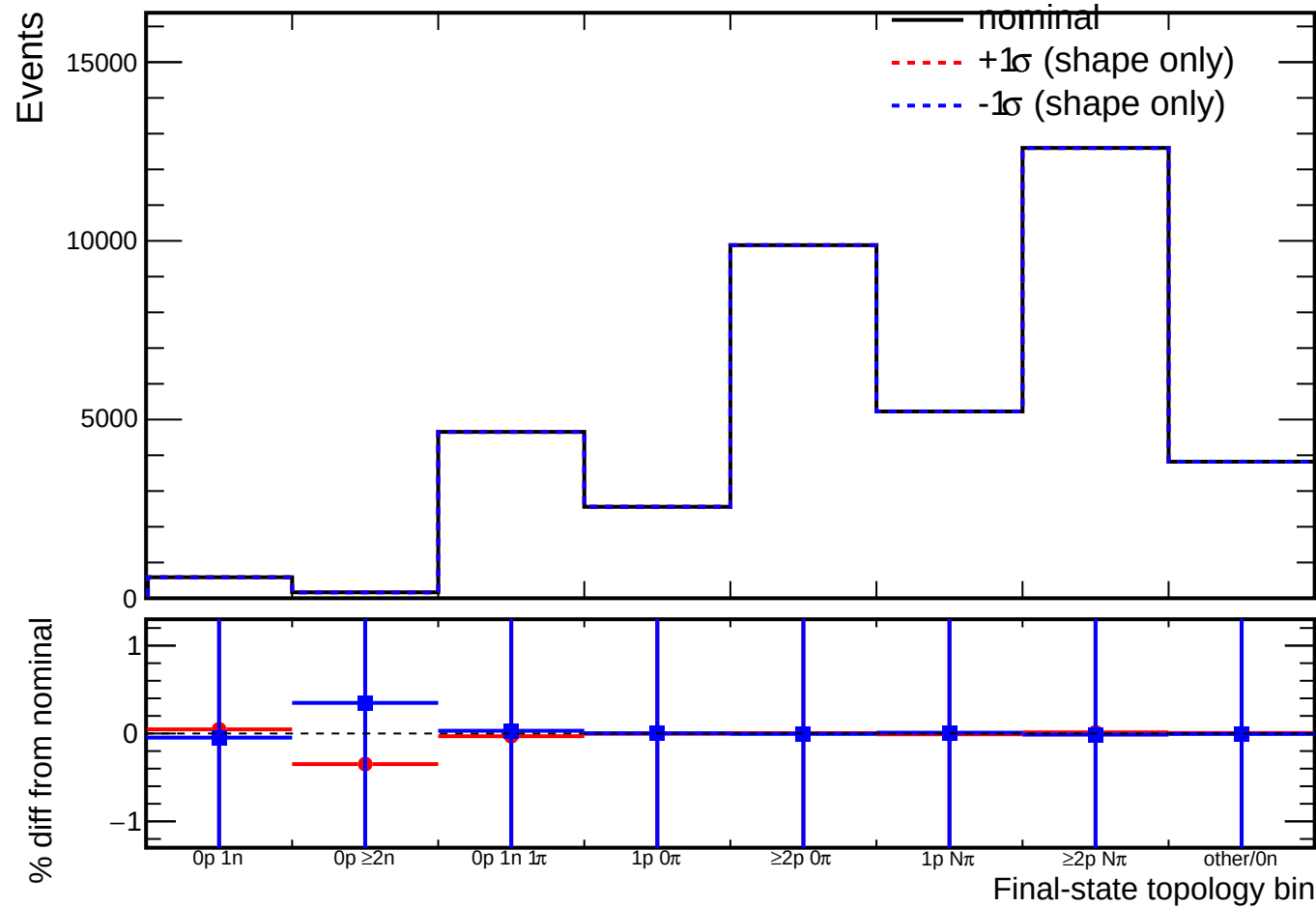
FHC [RES]: Theta_Delta2Npi (shape only) $\chi^2_{\text{shape},+1\sigma}=0.90$ $\chi^2_{\text{shape},-1\sigma}=0.40$ $\Delta\langle x\rangle_{+1\sigma}=+0.0036$ $\Delta\langle x\rangle_{-1\sigma}=-0.0024$
 $\chi^2_{\text{raw},+1\sigma}=3.45$ $\chi^2_{\text{raw},-1\sigma}=1.53$ rate: +0.80% / +0.53% $\Delta\sigma^2$: -0.0115 / -0.0077



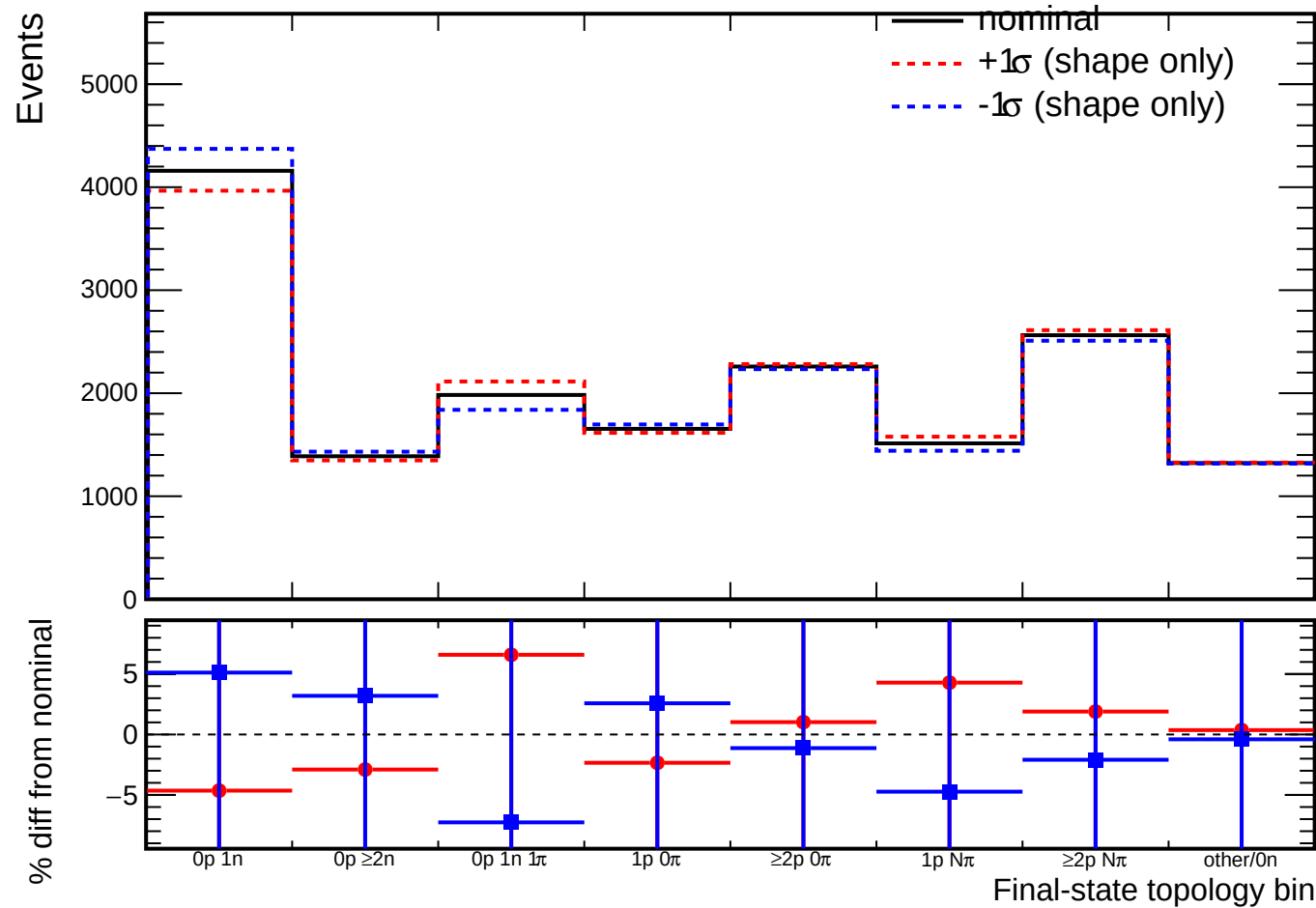
FHC [RES]: MvNCRES (shape only) $\chi^2_{\text{shape},+1\sigma}=0.83$ $\chi^2_{\text{shape},-1\sigma}=0.53$ $\Delta\langle x\rangle_{+1\sigma}=-0.0005$ $\Delta\langle x\rangle_{-1\sigma}=+0.0005$
 $\chi^2_{\text{raw},+1\sigma}=2.64$ $\chi^2_{\text{raw},-1\sigma}=1.63$ rate: +0.68% / -0.53% $\Delta\sigma^2$: +0.0061 / -0.0050



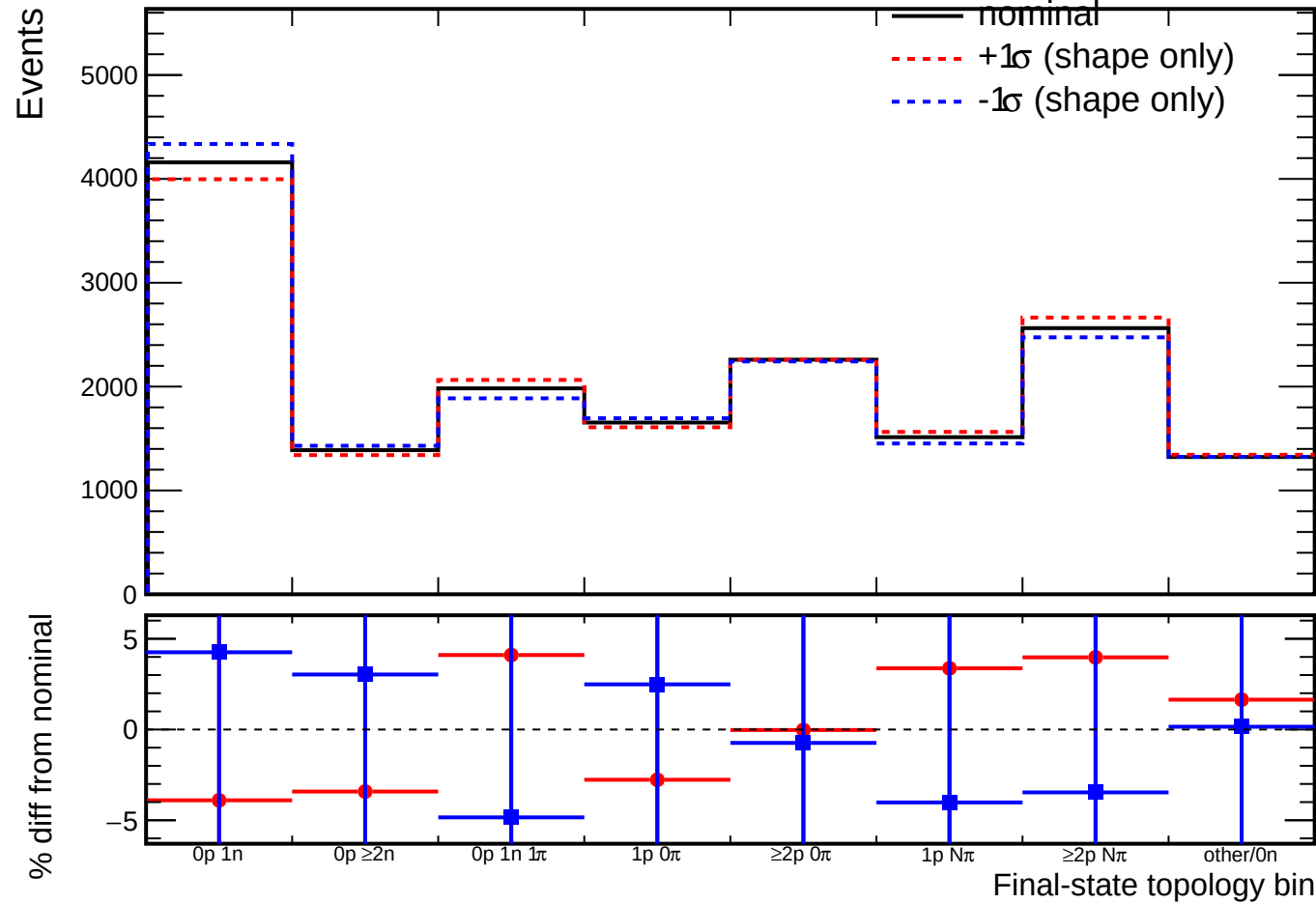
FHC [RES]: RESvpvnRatioNubarXSecSyst (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0002$ $\Delta\langle x\rangle_{-1\sigma}=-0.0002$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: -0.01% / +0.01% $\Delta\sigma^2$: -0.0002 / +0.0002



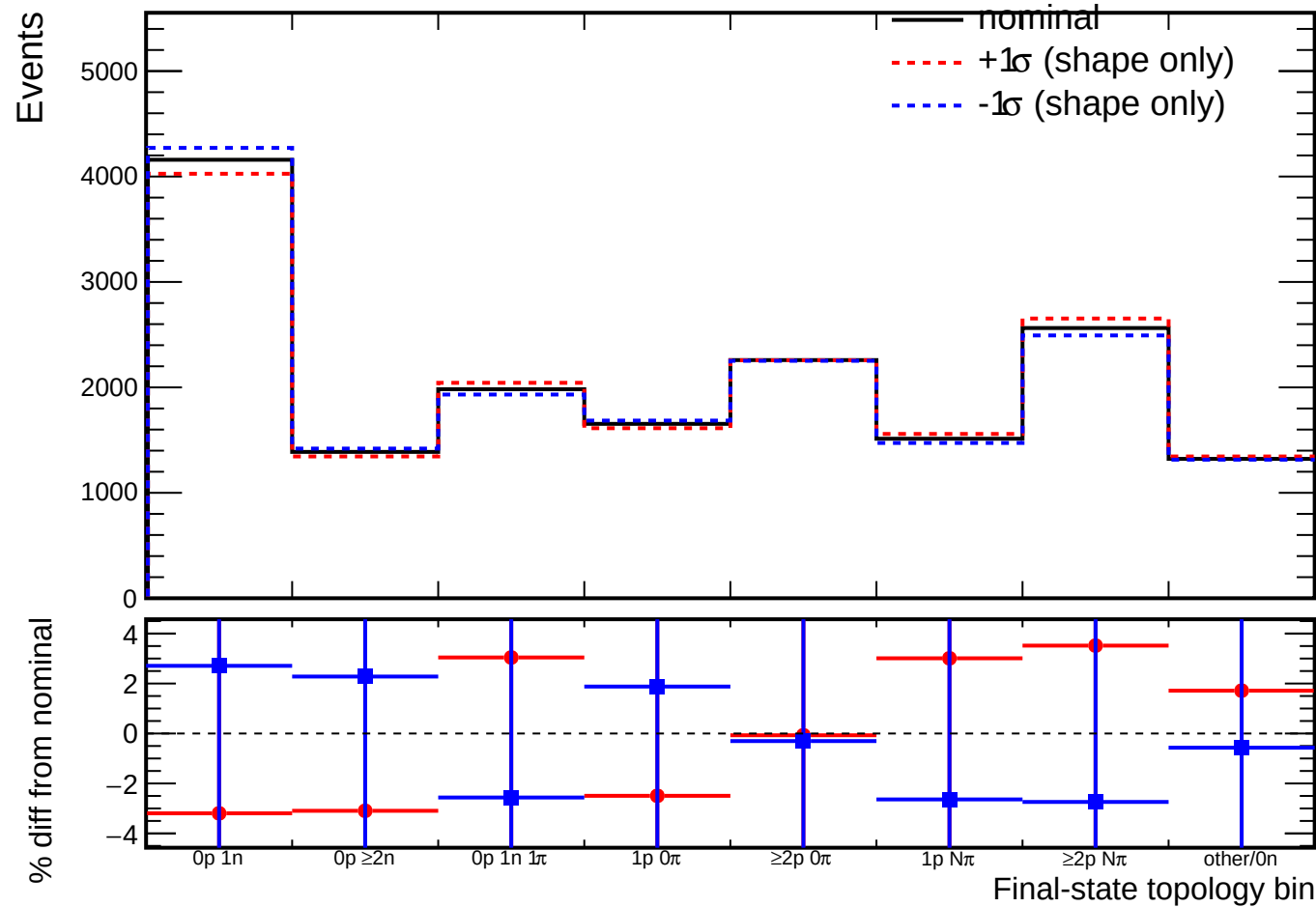
RHC [RES]: RESDeltaScaleSyst (shape only) $\chi^2_{\text{shape}, +1\sigma} = 23.62$ $\chi^2_{\text{shape}, -1\sigma} = 28.73$ $\Delta\langle x \rangle_{+1\sigma} = +0.0503$ $\Delta\langle x \rangle_{-1\sigma} = -0.0555$
 $\chi^2_{\text{raw}, +1\sigma} = 66.19$ $\chi^2_{\text{raw}, -1\sigma} = 66.19$ rate: +4.89% / -4.89% $\Delta\sigma^2$: -0.0664 / +0.0673



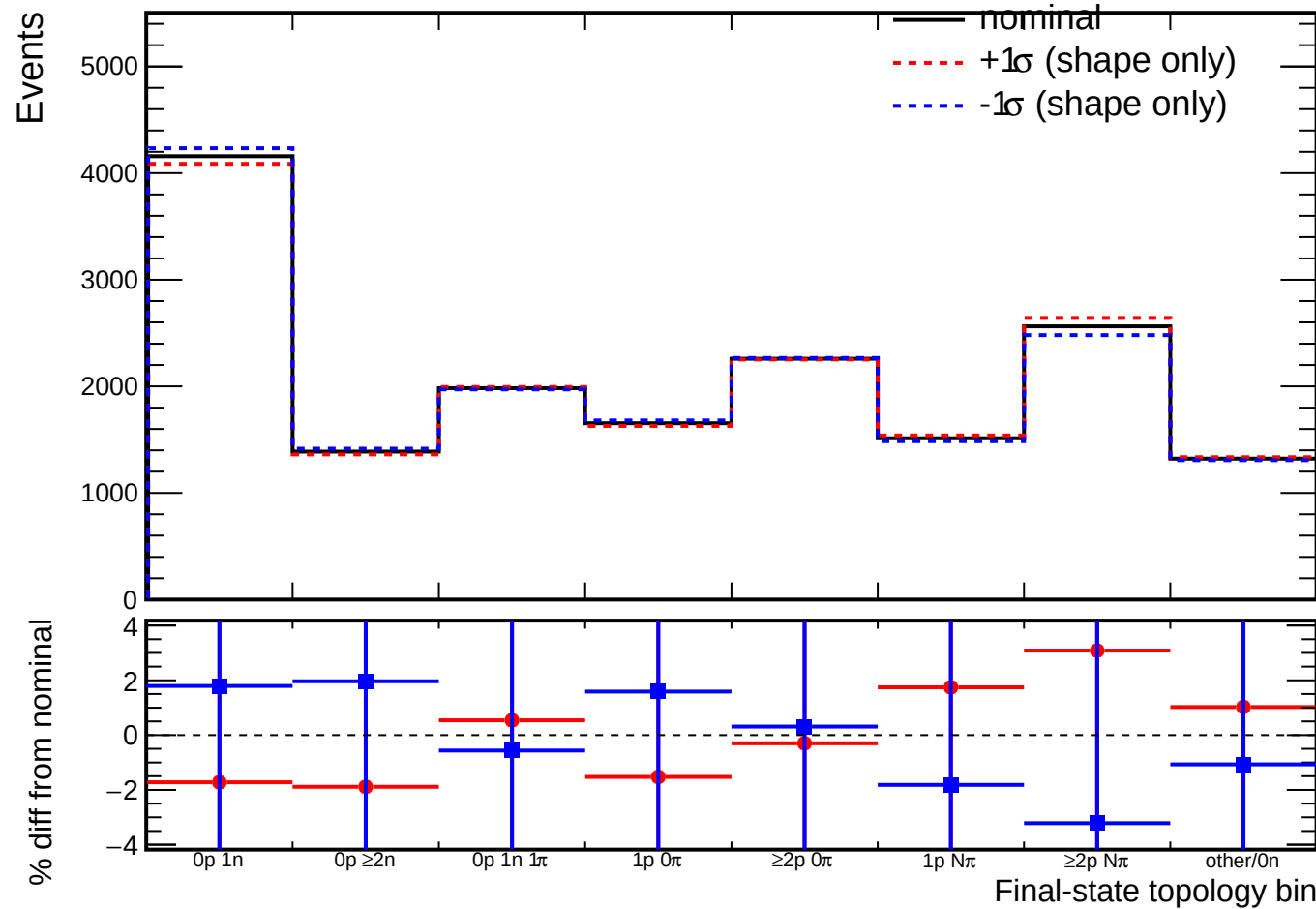
RHC [RES]: MaCCRES (shape only) $\chi^2_{\text{shape}, +1\sigma} = 18.69$ $\chi^2_{\text{shape}, -1\sigma} = 20.13$ $\Delta\langle x \rangle_{+1\sigma} = +0.0591$ $\Delta\langle x \rangle_{-1\sigma} = -0.0544$
 $\chi^2_{\text{raw}, +1\sigma} = 52.90$ $\chi^2_{\text{raw}, -1\sigma} = 44.93$ rate: +4.39% / -3.96% $\Delta\sigma^2$: -0.0166 / +0.0416



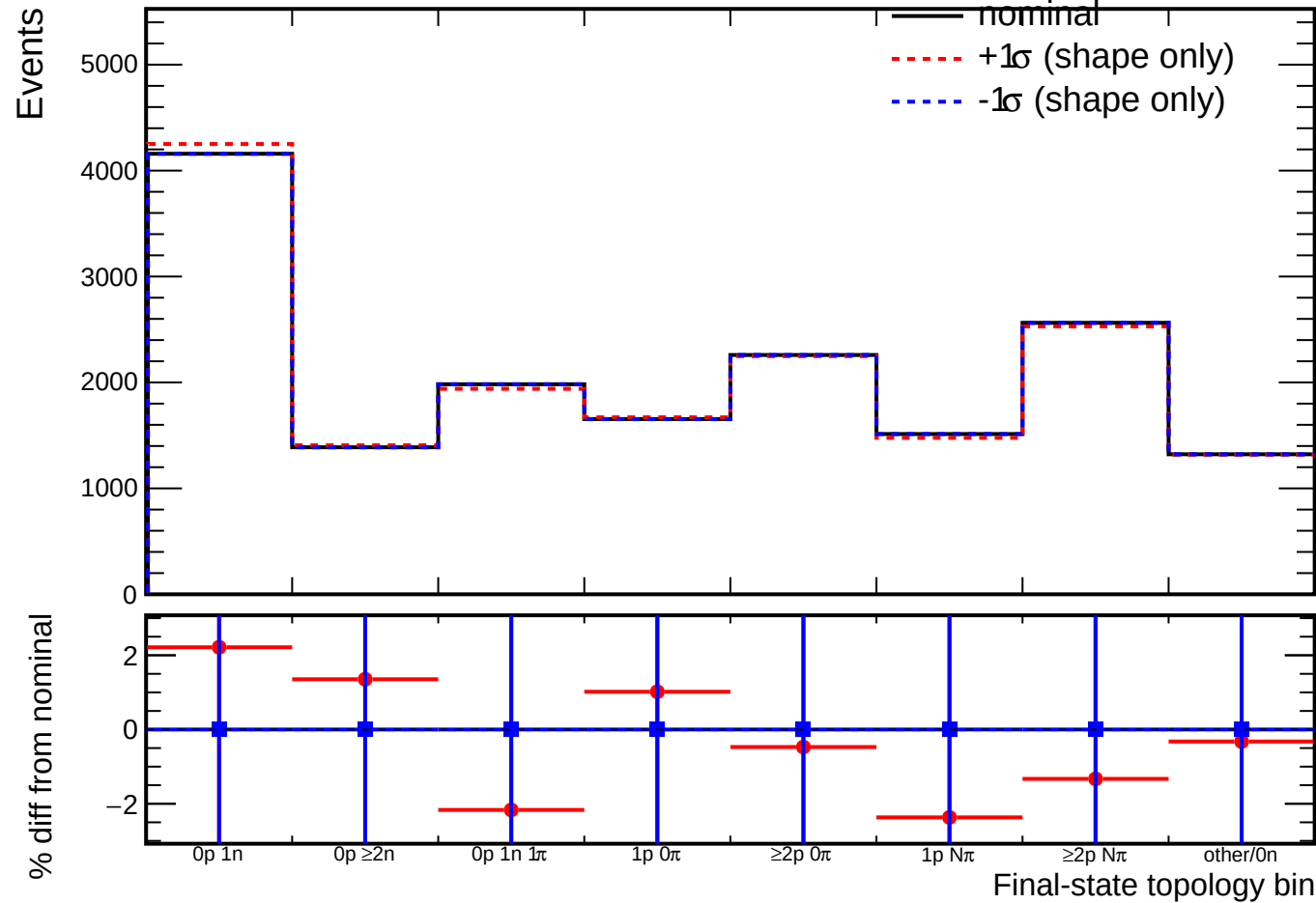
RHC [RES]: MvCCRES (shape only) $\chi^2_{\text{shape},+1\sigma}=13.37$ $\chi^2_{\text{shape},-1\sigma}=8.70$ $\Delta\langle x\rangle_{+1\sigma}=+0.0519$ $\Delta\langle x\rangle_{-1\sigma}=-0.0402$
 $\chi^2_{\text{raw},+1\sigma}=38.46$ $\chi^2_{\text{raw},-1\sigma}=20.63$ rate: +3.78% / -2.71% $\Delta\sigma^2$: -0.0062 / +0.0134



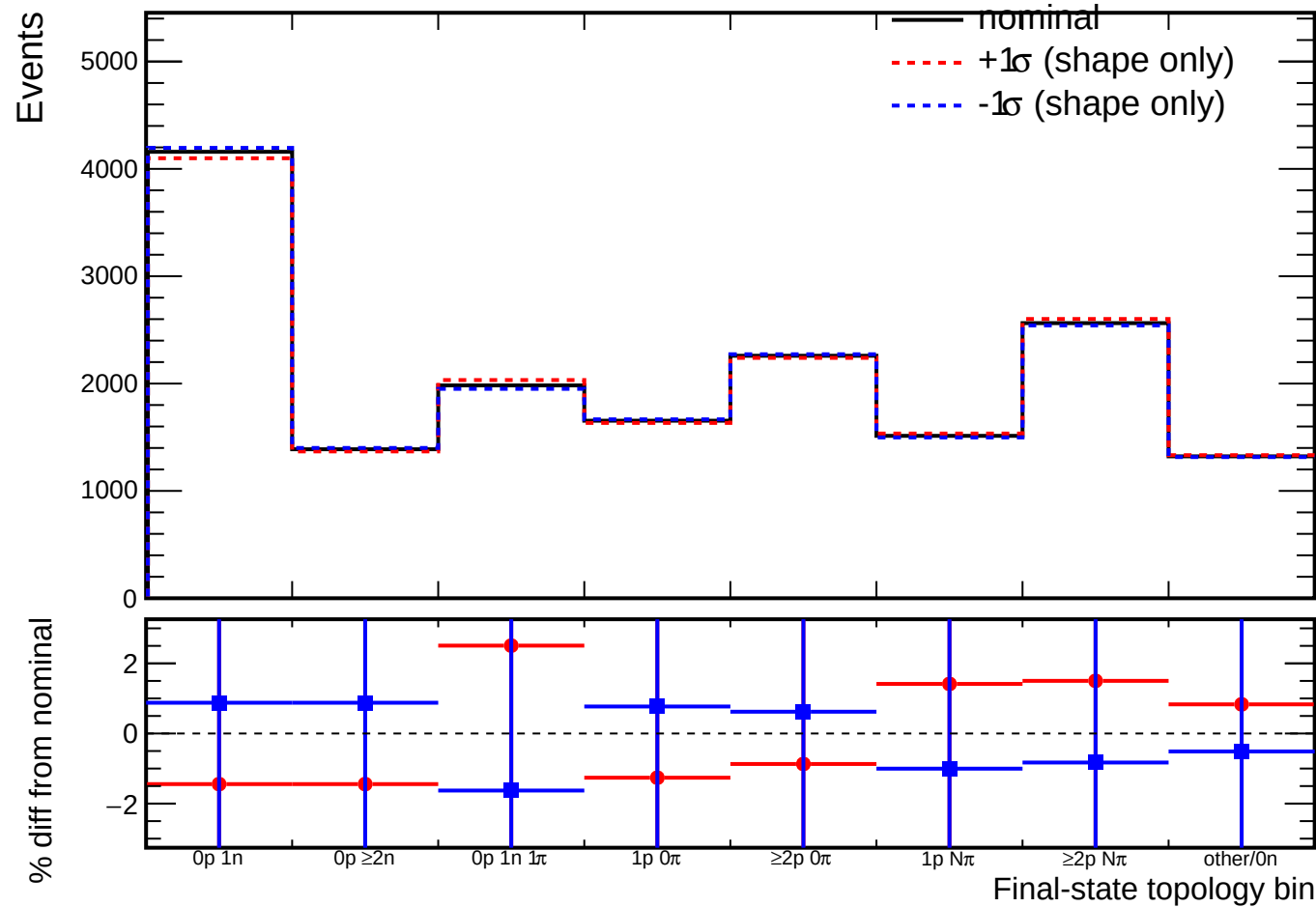
RHC [RES]: RESOtherScaleSyst (shape only) $\chi^2_{\text{shape},+1\sigma}=5.23$ $\chi^2_{\text{shape},-1\sigma}=5.67$ $\Delta\langle x\rangle_{+1\sigma}=+0.0353$ $\Delta\langle x\rangle_{-1\sigma}=-0.0367$
 $\chi^2_{\text{raw},+1\sigma}=12.46$ $\chi^2_{\text{raw},-1\sigma}=12.46$ rate: +2.04% / -2.04% $\Delta\sigma^2$: +0.0116 / -0.0148



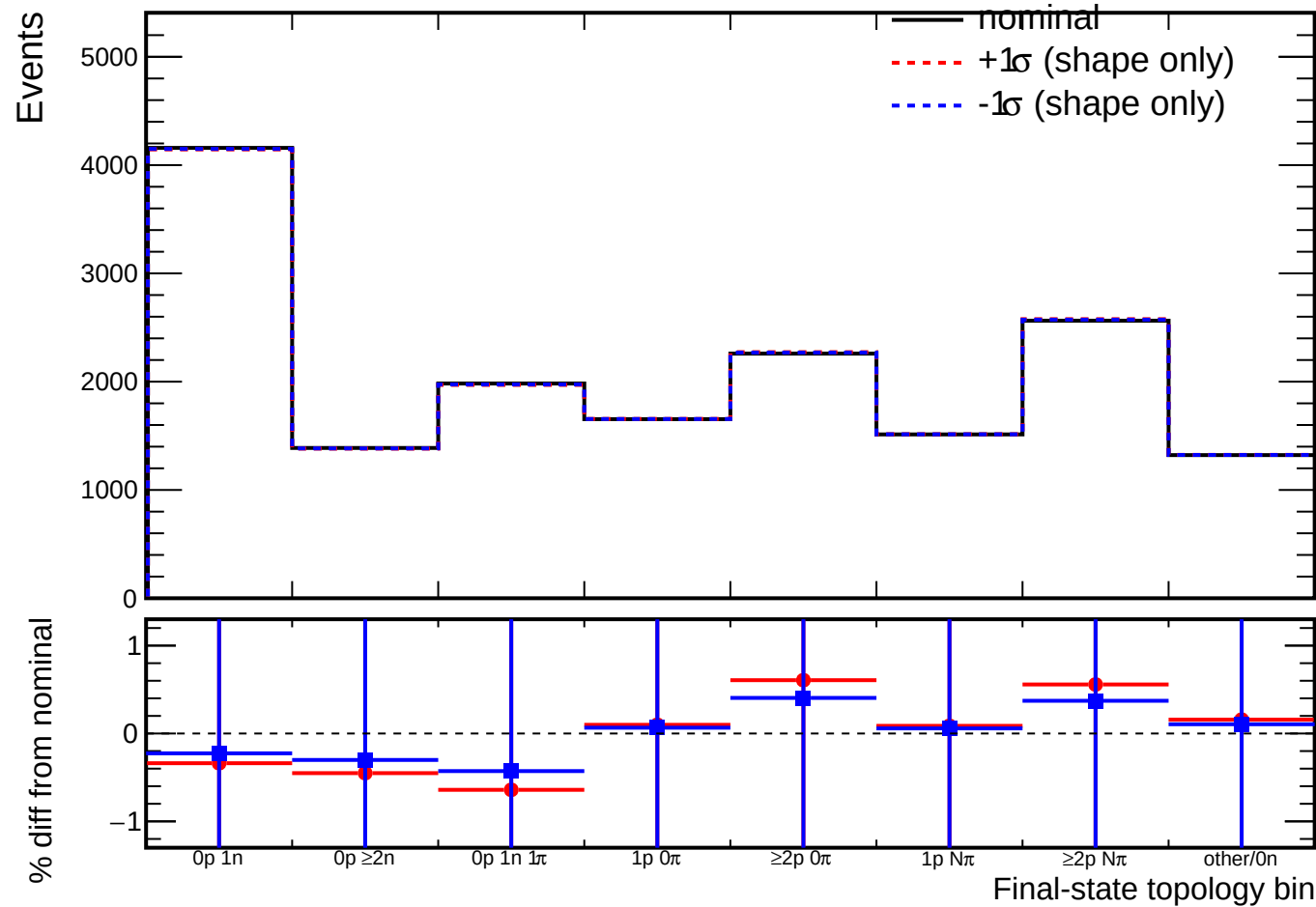
RHC [RES]: LowQ2RESSupp2020 (shape only) $\chi^2_{\text{shape},+1\sigma}=4.76$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x \rangle_{+1\sigma}=-0.0281$ $\Delta\langle x \rangle_{-1\sigma}=+0.0000$
 $\chi^2_{\text{raw},+1\sigma}=12.87$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: -2.22% / +0.00% $\Delta\sigma^2$: +0.0224 / +0.0000



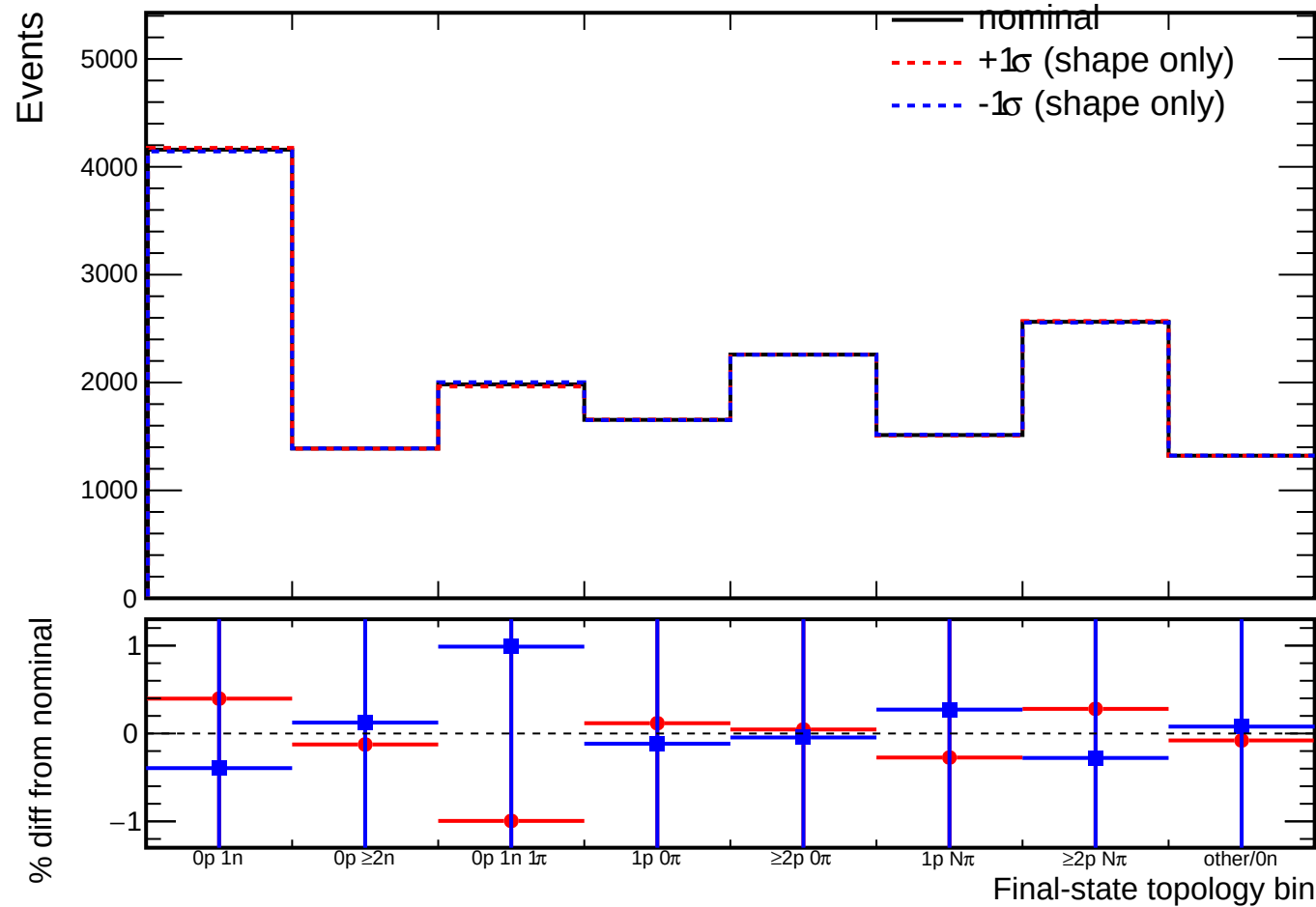
RHC [RES]: MaNCRES (shape only) $\chi^2_{\text{shape},+1\sigma}=3.81$ $\chi^2_{\text{shape},-1\sigma}=1.50$ $\Delta\langle x\rangle_{+1\sigma}=+0.0210$ $\Delta\langle x\rangle_{-1\sigma}=-0.0124$
 $\chi^2_{\text{raw},+1\sigma}=7.54$ $\chi^2_{\text{raw},-1\sigma}=2.75$ rate: +1.46% / -0.87% $\Delta\sigma^2$: -0.0020 / +0.0013



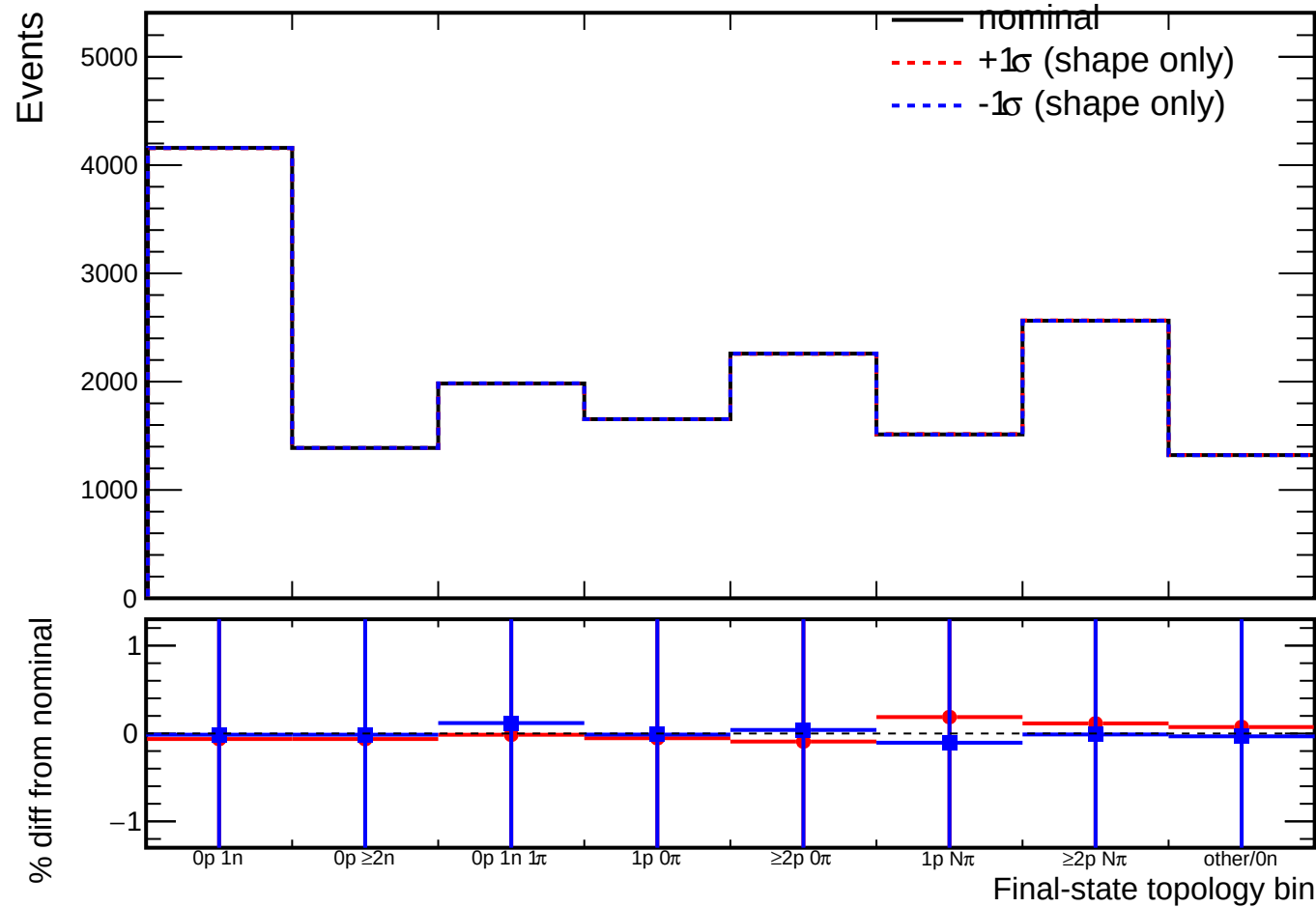
RHC [RES]: Theta_Delta2Npi (shape only) $\chi^2_{\text{shape},+1\sigma}=0.33$ $\chi^2_{\text{shape},-1\sigma}=0.15$ $\Delta\langle x\rangle_{+1\sigma}=+0.0080$ $\Delta\langle x\rangle_{-1\sigma}=+0.0053$
 $\chi^2_{\text{raw},+1\sigma}=0.52$ $\chi^2_{\text{raw},-1\sigma}=0.23$ rate: +0.34% / +0.23% $\Delta\sigma^2$: -0.0001 / -0.0000



RHC [RES]: RESvpvnRatioNubarXSecSyst (shape only) $\chi^2_{\text{shape}, +1\sigma} = 0.30$ $\chi^2_{\text{shape}, -1\sigma} = 0.30$ $\Delta\langle x \rangle_{+1\sigma} = -0.0010$ $\Delta\langle x \rangle_{-1\sigma} = +0.0010$
 $\chi^2_{\text{raw}, +1\sigma} = 0.47$ $\chi^2_{\text{raw}, -1\sigma} = 0.47$ rate: -0.32% / +0.32% $\Delta\sigma^2$: +0.0093 / -0.0092



RHC [RES]: MvNCRES (shape only) $\chi^2_{\text{shape},+1\sigma}=0.01$ $\chi^2_{\text{shape},-1\sigma}=0.01$ $\Delta\langle x\rangle_{+1\sigma}=+0.0016$ $\Delta\langle x\rangle_{-1\sigma}=-0.0003$
 $\chi^2_{\text{raw},+1\sigma}=0.02$ $\chi^2_{\text{raw},-1\sigma}=0.01$ rate: +0.06% / +0.01% $\Delta\sigma^2$: +0.0012 / -0.0011



RHC [RES]: RESvpvnRatioNuXSecSyst (shape only) $\chi^2_{\text{shape},+1\sigma}=0.00$ $\chi^2_{\text{shape},-1\sigma}=0.00$ $\Delta\langle x\rangle_{+1\sigma}=+0.0001$ $\Delta\langle x\rangle_{-1\sigma}=-0.0001$
 $\chi^2_{\text{raw},+1\sigma}=0.00$ $\chi^2_{\text{raw},-1\sigma}=0.00$ rate: -0.01% / +0.01% $\Delta\sigma^2$: -0.0001 / +0.0001

